MATHEMATICS

DETERMINANT
THEORY SHEET
ENGLISH

MATRIX

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1.1 INTRODUCTION OF DETERMINANT

An expression expressed in equal number of rows and column and put between two vertical lines is named as determinant of that expression e.g. $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ are the determinants of second and third order respectively.

Number of rows of a determinant is equal to number of columns and it is equal to order of determinant.

$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ Element $a_{ij} \Rightarrow$ i = position of row j = position of column

1.2 VALUE OF A DETERMINANT BY EXPANSION

$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is a determinant of second order. These numbers are called the elements of the determinant. The value of the determinant is defined as

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1$$

$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is determinant of third order.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

The diagonal through the left-hand top corner which contains the element a_1, b_2, c_3 is called the leading diagonal or principal diagonal

Note: We can expand det. along any row or column. For easier calculation purpose, we will expand det. along that particular row or column which contain maximum number of zeroes.

1.3 MINOR AND COFACTOR OF ELEMENTS OF A DETERMINANT

The minor of a given element of determinant is the determinant obtained by deleting the row & the column in which the given element stands.

For example, the minor of a_1 in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order three will have "9 minors".

If M_{ij} represents the minor of the element belonging to i^{th} row and j^{th} column then the cofactor of that element is given by : $C_{ij} = (-1)^{i+j} . M_{ij}$

Note:

1. By multiplying elements of a row with corresponding cofactors and then added, resultant sum will be equal to the value of det.

$$D=\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Expand along R_1 $\quad D = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

Expand along C_2 $\quad D = a_{12}M_{12} + a_{22}M_{22} - a_{32}M_{32} = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}$

2. On multiplying elements of any row with co-factor of corresponding elements of other row then added then the resultant sum will be zero.

$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$

Q.1 Find x

(i) $x=\begin{vmatrix} 2 & 0 & 4 \\ 3 & 1 & -2 \\ 4 & 1 & 0 \end{vmatrix}$ (ii) $\begin{vmatrix} x & 4 & 2 \\ 2x & 1 & 4 \\ 0 & 3 & 2 \end{vmatrix}=0$ (iii) $\begin{vmatrix} x & 2 & 4 \\ -1 & 1 & 1 \\ x+1 & 1 & 3 \end{vmatrix}=0$

Ans. (i) $x=0$ (ii) $x=0$ (iii) $x \in R$

Q.2 Find the minors and cofactors of elements '–3', '5', '–1' & '7' in the determinant $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$

Ans (33, –33), (9, –9), (–15, –15), (12,12)

Sol. Minor of $-3=\begin{vmatrix} 4 & 5 \\ -1 & 7 \end{vmatrix}=33$; Cofactor of $-3=-33$

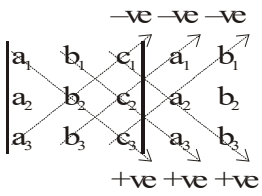
Minor of $5=\begin{vmatrix} 2 & -3 \\ -1 & 6 \end{vmatrix}=9$; Cofactor of $5=-9$

Minor of $-1=\begin{vmatrix} -3 & 1 \\ 0 & 5 \end{vmatrix}=-15$; Cofactor of $-1=-15$

Minor of $7=\begin{vmatrix} 2 & -3 \\ 4 & 0 \end{vmatrix}=12$; Cofactor of $7=12$

1.4 SARRUS RULE

$$D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=$$



$$= (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$$

2.1 PROPERTIES OF DETERMINANT

P-1 The value of Determinant remains unchanged, if the rows and the column are interchanged. This is always denoted by ' and is also called transpose

e.g. $D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_2 & b_3 & c_3 \end{vmatrix}$ and $D'=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

Then D' = D, D and D' are transpose of each other

Note :

Since the Determinant remains unchanged when rows and columns are interchanged, it is obvious that any theorem which is true for 'rows' must also be true for 'Columns'

P-2 If any two rows (or columns) of a determinant be interchanged, the determinant is unaltered in numerical Value, but is changed in sign only,

e.g. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ Then $D' = -D$

P-3 If a Determinant has two rows (or columns) identical, then its value is zero.

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$ Then, $D = 0$

P-4 If all the elements of any row (or column) be multiplied by the same number, then the value of Determinant is multiplied by that number.

e.g. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} ka_1 & kb_1 & kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ Then $D' = kD$

Note:

1. If all the elements of a det. are multiplied by k then value of that det. gets multiplied by k^n where n is the order of det.

$$\begin{vmatrix} ka_1 & ka_2 & ka_3 \\ kb_1 & kb_2 & kb_3 \\ kc_1 & kc_2 & kc_3 \end{vmatrix} = k^3 \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

2. If elements of any two rows of det. are proportional then value of det. will be zero.

for example : $\begin{vmatrix} ka_1 & kb_1 & kc_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

P-5 If each element of any row (or column) can be expressed as a sum of two terms, then the determinant can be expressed as the sum of the Determinants

e.g. $\begin{vmatrix} a_1+x & b_1+y & c_1+z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

P-6 If $\Delta = f(x)$ and $f(a) = 0$ then $(x-a)$ is a factor of Δ

e.g. $D = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

if we replace a by b then

$$D = \begin{vmatrix} 1 & 1 & 1 \\ b & b & c \\ b^2 & b^2 & c^2 \end{vmatrix} = 0 \Rightarrow (a - b) \text{ is a factor of } D$$

2.2 ROW AND COLUMN TRANSFORMATION

1 **Row - column operation :** The value of a determinant remains unaltered under a column (C_i) operation of the form $C_i \to C_i + \alpha C_j + \beta C_k$ (j, k ≠ i) or row (R_i) operation of the form $R_i \to R_i + \alpha R_j + \beta R_k$ (j, k ≠ i). In other words, the value of a determinant is not altered by adding the elements of any row (or column) to the same multiples of the corresponding elements of any other row (or column)

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$D = \begin{vmatrix} a_1 + \alpha a_2 & b_1 + \alpha b_2 & c_1 + \alpha c_2 \\ a_2 & b_2 & c_2 \\ a_3 + \beta a_1 & b_3 + \beta b_1 & c_3 + \beta c_1 \end{vmatrix} (R_1 \to R_1 + \alpha R_2; R_3 \to R_3 + \beta R_2)$$

Note : By using the operation $R_i \to xR_i + yR_j + zR_k$ (j, k ≠ i), the value of the determinant becomes x times the original one.

2.3 FOUR STANDARD DETERMINANTS

1. $$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc) = -(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

Ans.

Sol. $$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\Rightarrow \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}$$

Take common $(b - a)$ from R_2

Take common $(c - a)$ from R_3

$$D = (b-a)(c-a)\begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{vmatrix}$$

rxpand along C_1

$$D=(b-a)(c-a)\left[1\begin{vmatrix}1, & b+a\\ & 1\end{vmatrix}-0\,||+0\,||\right]$$

$D = (b - a)\,(c - a)\,(c + a - b - a)$

$D = (b - a)\,(c - a)\,(c - b)$

$= (a - b)\,(b - c)\,(c - a)$

2. $\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}=(a-b)(b-c)(c-a)$

Ans.

Sol. $D=\begin{vmatrix}1 & a & a^3\\ 1 & b & b^3\\ 1 & c & c^3\end{vmatrix}$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$D=\begin{vmatrix}1 & a & a^3\\ 0 & b-a & b^3-a^3\\ 0 & c-a & c^3-a^3\end{vmatrix}$$

$$D=\begin{vmatrix}1 & a & a^3\\ 0 & b-a & (b-a)\left(b^2+c^2+ab\right)\\ 0 & c-a & (c-a)\left(c^2+a^2+ca\right)\end{vmatrix}$$

Take common $(b - a)$ from R_1

Take common $(c - a)$ from R_2

$$D=(b-a)(c-a)\begin{vmatrix}1 & a & a^3\\ 0 & 1 & b^2+c^2+ab\\ 0 & 1 & c^2+a^2+ac\end{vmatrix}$$

rxpand along C_1

$$D=(b-a)(c-a)\left\{1\begin{vmatrix}1 & b^2+a^2+ab\\ 1 & c^2+a^2+ac\end{vmatrix}-|\,0\,|+|\,0\,|\right\}$$

$D = (b - a)\,(c - a)\,(c^2 + a^2 + ac - b^2 - a^2 - ab)$

$\Rightarrow (b - a)\,(c - a)\,[(c - b)(c + b) + a(c - b)]$

$\Rightarrow (b-a)(c-a)(c-b)(a+b+c)$

$\Rightarrow (a-b)(b-c)(c-a)(a+b+c)$

3. $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$

Ans.

Sol. $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$\Rightarrow \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2-a^2 & b^3-a^3 \\ 0 & c^2-a^2 & c^3-a^3 \end{vmatrix}$ rxpand along C_1

$(b-a)(c-a)\left\{1\begin{vmatrix} b+a & b^2+a^2+ab \\ c+a & c^2+a^2+ac \end{vmatrix} - 0 + 0\right\}$

$\Rightarrow (b-a)(c-a)(bc^2 + ba^2 + abc + ac^2 + a^3 + a^2c - cb^2 - ca^2 - abc - ab^2 - a^3 - a^2b)$

$\Rightarrow (b-a)(c-a)\,(bc\,(c-b) + a(c^2 - b^2))$

$\Rightarrow (b-a)(c-a)\,[bc\,(c-b) + a(c+b) + (c-b)]$

$\Rightarrow (b-a)(c-a)(c-b)\,[bc + ac + ab]$

$\Rightarrow (a-b)\,(b-c)\,(c-a)\,[ab + bc + ac]$

4. $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

Ans.

Sol. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ $R_1 \to R_1 + R_2 + R_3$

$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix}$ Take $a + b + c$ common from R_1

$$\Rightarrow (a+b+c)\begin{vmatrix}1 & 1 & 1\\ b & c & a\\ c & a & b\end{vmatrix}$$

$C_2 \to C_2 - C_1$

$C_3 \to C_3 - C_1$

$$\Rightarrow (a+b+c)\begin{vmatrix}1 & 0 & 0\\ b & c-b & a-b\\ c & a-c & b-c\end{vmatrix}$$ Expand along R_1

$$\Rightarrow (a+b+c)\left\{1\begin{vmatrix}c-b & a-b\\ a-c & b-c\end{vmatrix}-0+0\right\}$$

$\Rightarrow (a + b + c)(bc - b^2 - c^2 + bc - a^2 + ab + ac - bc)$

$\Rightarrow -(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$

$\Rightarrow -(a^3 + b^3 + c^3 - 3abc)$

2.4 TWO IMPORTANT OPERATIONS

1. If all the elements of any row/column are one.

$$\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix} \qquad \begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}$$

$R_1 \to R_1 - R_3 ; R_2 \to R_2 - R_3$ or $C_1 \to C_1 - C_3 ; C_2 \to C_2 - C_3$

Expand along C_1 Expand along R_1

2. If on adding all three rows or columns, a common factor gets formed

$$\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix} \qquad R_1 \to R_1 + R_2 + R_3 \text{ or } C_1 \to C_1 + C_2 + C_3$$

Q.1 If $\begin{vmatrix}x+1 & 2x & 3\\ 2x & 3 & x+1\\ 3 & x+1 & 2x\end{vmatrix} = 0$ find x

Ans. $x = -\frac{4}{3}$

Sol. $$\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix} = -\left(a^3 + b^3 + c^3 - 3abc\right)$$

By standard determinant

$\Rightarrow -\{(x + 1)^3 + (2x)^3 + 3^3 - 3(x + 1)(2x)(3)\} = 0$

So that

$x + 1 + 2x + 3 = 0 \qquad x + 1 = ax = 3$

$3x + 4 = 0$

$$x = -\frac{4}{3}$$

Q.2 Prove that $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Ans.

Sol. By Standard

$\Rightarrow -\{(a + b)^3 + (b + c)^3 + (c + a)^3 - 3(a + b)(b + c)(c + a)\}$

$\Rightarrow -\{a^3 + b^3 + 3ab^2 + 3a^2b + b^3 + c^3 + 3bc^2 + 3b^2c + a^3 + c^3 + 3ac^2 + 3a^2c$

$- 3 (abc + ac^2 + b^2c + bc^2 + a^2b + a^2c + b^2a + abc)\}$

$\Rightarrow -\{2a^3 + 2b^3 + 2c^3 - 6abc\}$

$\Rightarrow -2(a^3 + b^3 + c^3 - 3abc)$

$$\Rightarrow 2\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Q.3 Prove that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Ans.

Sol. $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Take common a from C_1

Take common b from C_2

Take common c from C_3

$$\Rightarrow abc\begin{vmatrix} 1+\frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & 1+\frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix} \qquad C_1 \to C_1 + C_2 + C_3$$

$$\Rightarrow abc\begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & \frac{1}{b} & \frac{1}{c} \\ 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{b} & \frac{1}{c} \\ 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix}$$

Take common $1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ from C_1.

$$\Rightarrow (abc)\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 1 & 1+\frac{1}{b} & \frac{1}{c} \\ 1 & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix}$$

$R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$$\Rightarrow (abc)\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expand along C_1

$$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$

Q.4 Prove that $\begin{vmatrix} 23 & 66 & 11 \\ 36 & 55 & 26 \\ 63 & 143 & 37 \end{vmatrix} = 0$

Ans.

Sol. Take common 11 from C_2

$$11\begin{vmatrix} 23 & 6 & 11 \\ 36 & 5 & 26 \\ 63 & 13 & 37 \end{vmatrix}$$

$C_1 \to C_1 - C_3$

$$11\begin{vmatrix} 12 & 6 & 11 \\ 10 & 5 & 26 \\ 26 & 13 & 37 \end{vmatrix}$$

Take common 2 from C_1

$$22\begin{vmatrix} 6 & 6 & 11 \\ 5 & 5 & 26 \\ 13 & 13 & 37 \end{vmatrix} = 0$$ (Hence prowed)

Q.5 Prove that $\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ac \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a^2 + b^2 + c^2)(a+b+c)(a-b)(b-c)(c-a)$

Ans.

Sol. $\begin{vmatrix} b^2+c^2+2bc & a^2 & bc \\ c^2+a^2+2ac & b^2 & ac \\ a^2+b^2+2ab & c^2 & ab \end{vmatrix}$

$C_1 \to C_1 + C_2 - 2C_3$

$\Rightarrow \begin{vmatrix} a^2+b^2+c^2 & a^2 & bc \\ a^2+b^2+c^2 & b^2 & ac \\ a^2+b^2+c^2 & c^2 & ab \end{vmatrix}$

$\Rightarrow \left(a^2+b^2+c^2\right)\cdot \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ac \\ 1 & c^2 & ab \end{vmatrix}$

Multiply R_1 by a
Multiply R_2 by b
Multiply R_3 by c

$\Rightarrow \frac{a^2+b^2+c^2}{abc}\begin{vmatrix} a & a^3 & abc \\ b & b^3 & abc \\ c & c^3 & abc \end{vmatrix}$

$\Rightarrow \frac{\left(a^2+b^2+c^2\right)abc}{abc}\begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix}$

$\Rightarrow \left(a^2+b^2+c^2\right)(a-b)(b-c)(c-a)(a+b+c)$

Q.6 If $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ Prove that $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$

Ans.

Sol. $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$

$R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$\Rightarrow \begin{vmatrix} p & b & c \\ a-p & q-b & o \\ a-p & o & r-c \end{vmatrix} = 0$

$P(q-b)(r-c) - b(a-p)(r-c) - c(a-p)(q-b) = 0$
$P(q-b)(r-c) + b(p-a)(r-c) + c(p-a)(q-b) = 0$

$\div (p-a)(q-b)(r-c)$

$$\Rightarrow \frac{P}{p-a}+\frac{b}{q-b}+\frac{c}{r-c}=0$$

$$\Rightarrow \frac{P}{p-a}+\frac{b}{q-b}+1+\frac{c}{r-c}+1=2$$

$$\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$$

Q.7 The solutions of the equation $\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$, $(0 < x < \pi)$, are :

[JEE-Main,March-2021]

समीकरण $\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$, $(0 < x < \pi)$ के हल है : **[JEE-Main,March-2021]**

Ans. $\frac{7\pi}{12}, \frac{11\pi}{12}$

Sol. $R_1 \to R_1 + R_2$

$$\Rightarrow \Delta = \begin{vmatrix} 2 & 2 & 1 \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin^2 x & 4\sin^2 x & 1+4\sin^2 x \end{vmatrix}$$

$C_2 \to C_2 - C_1$: $C_3 \to C_3 - C_1$

$$\Rightarrow \Delta \begin{vmatrix} 2 & 2 & -1 \\ \cos^2 x & 1 & 0 \\ 4\sin^2 x & 0 & 1 \end{vmatrix} = 0$$

$\Rightarrow 2 + 4\sin 2x = 0$

$$\Rightarrow \sin 2x = -\frac{1}{2}$$

$$\Rightarrow 2x = \frac{7\pi}{6}.\frac{11\pi}{6} \Rightarrow x = \frac{7\pi}{12}, \frac{11\pi}{12}$$

Q.8 If three digit numbers A38, 3B9, 62C are divisible by k then prove that $\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 3 & B & 2 \end{vmatrix}$ is also divisible by k.

Ans.

Sol. $A38 = 100A + 3 \times 10 + 8 = KI_1$

$AB8 = 300 + B \times 10 + 9 = KI_2$

$62C = 600 + 20 + C = KI_3$

$$\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 3 & B & 2 \end{vmatrix} \qquad R_2 \to 100R_1 + 10 \times R_3 + R_2$$

$$= \begin{vmatrix} A & 3 & 6 \\ 100A + 30 + 8 & 300 + B \times 10 + 9 & 600 + 2 \times 10 + C \\ 3 & B & 2 \end{vmatrix}$$

$$= \begin{vmatrix} A & 3 & 6 \\ KI_1 & KI_2 & KI_3 \\ 3 & B & 2 \end{vmatrix}$$

$$= K \begin{vmatrix} A & 3 & 6 \\ I_1 & I_2 & I_3 \\ 3 & B & 2 \end{vmatrix}$$

Q.9 $\begin{vmatrix} (b+c)^2 & ab & ac \\ ab & (a+c)^2 & bc \\ ac & bc & (a+b)^2 \end{vmatrix} = 2\,(abc)\,(a+b+c)^3.$

Sol. Multiply C_1 by a

Multiply C_2 by b

Multiply C_3 by c

$$\Rightarrow \frac{1}{abc} \begin{vmatrix} a(b+c)^2 & ab^2 & ac^2 \\ a^2b & b(a+c)^2 & bc^2 \\ a^2c & b^2c & c(a+b)^2 \end{vmatrix}$$

Take common a from R_1

Take common b from R_2

Take common c from R_3

$$\Rightarrow \frac{abc}{abc}\begin{vmatrix}(b+c)^2 & b^2 & c^2\\ a^2 & (a+c)^2 & c^2\\ a^2 & b^2 & (a+b)^2\end{vmatrix}$$

$R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$$\Rightarrow \begin{vmatrix}(b+c)^2 & b^2 & c^2\\ a^2-(b+c)^2 & (a+c)^2-b^2 & 0\\ a^2-(b+c)^2 & 0 & (a+b)^2-c^2\end{vmatrix}$$

Take common (a + b + c) from R_2 and R_3

$$\Rightarrow (a+b+c)^2\begin{vmatrix}(b+c)^2 & b^2 & c^2\\ a-b-c & a+c-b & 0\\ a-b-c & 0 & a+b-c\end{vmatrix}$$

$C_1 \to C_1 - C_2 - C_3$

$$(a+b+c)^2\begin{vmatrix}2bc & b^2 & c^2\\ -2c & a+c-b & 0\\ -2b & 0 & a+b-c\end{vmatrix}$$

$\Rightarrow (a + b + c)^2 [2bc ((a + c – b)(a + b – c)) – b^2 (–2c (a + b – c)) + c^2 (2b(a + c – b))]$
$\Rightarrow (a + b + c)^2 (2bc (a^2 – (b – c)^2) + 2b^2c (a + b – c) + 2bc^2 (a + c – b)$
$2bc (a + b + c)2 [a^2 – (b – c)^2 + ab + b^2 – bc + ac + c^2 – bc]$
$2bc(a + b + c)^2 (a^2 – b^2 – c^2 + 2bc + ab + b^2 – bc + ac + c^2 – bc)$
$2bc(a + b + c)^2 (a^2 + ab + ac)$
$= 2abc(a + b + c)^2 (a + b + c)$
$= 2abc(a + b + c)^3$

MULTIPLICATION OF DETERMINANTS

Multiplication of two second order determinants is defined as follows

$$\begin{vmatrix}a_1 & b_1\\ a_2 & b_2\end{vmatrix} \times \begin{vmatrix}\ell_1 & m_1\\ \ell_2 & m_2\end{vmatrix}$$

$$= \begin{vmatrix}a_1\ell_1+b_1\ell_2 & a_1m_1+b_1m_2\\ a_2\ell_1+b_2\ell_2 & a_2m_1+b_2m_2\end{vmatrix}$$

Multiplication of two third order determinants is defined as follows

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1\ell_1 + b_1\ell_2 + c_1\ell_3 & a_1m_1 + b_1m_2 + c_1m_3 & a_1n_1 + b_1n_2 + c_1n_3 \\ a_2\ell_1 + b_2\ell_2 + c_2\ell_3 & a_2m_1 + b_2m_2 + c_2m_3 & a_2n_1 + b_2n_2 + c_2n_3 \\ a_3\ell_1 + b_3\ell_2 + c_3\ell_3 & a_3m_1 + b_3m_2 + c_3m_3 & a_3n_1 + b_3n_2 + c_3n_3 \end{vmatrix}$$

Here we have multiplied row by column. We can also multiply row by row, column by row and column by column.

Q.1 If $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$. Prove that $D^2 = \begin{vmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix}$ where c_{ij} is cofactor of a_{ij}.

Sol. Let $x = \begin{vmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix}$

$$D.X = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \times \begin{vmatrix} c_{11} & c_{12} & c_{33} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix}$$

Row × Row multiply

$$DX = \begin{vmatrix} a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} & a_{11}a_{21} + a_{12}c_{22} + a_{13}c_{23} & a_{11}a_{31} + a_{12}a_{22} + a_{13}a_{33} \\ a_{21}c_{11} + a_{22}c_{12} + a_{23}c_{13} & a_{21}c_{21} + a_{22}c_{22} + a_{23}c_{23} & a_{21}c_{31} + a_{22}c_{32} + c_{22}c_{23} \\ a_{31}c_{11} + a_{32}c_{12} + a_{33}c_{13} & a_{31}c_{21} + a_{32}c_{22} + a_{33}c_{23} & a_{31}c_{31} + a_{32}c_{33} + c_{33}c_{33} \end{vmatrix}$$

$$DX = \begin{vmatrix} D & O & O \\ O & D & O \\ O & O & D \end{vmatrix}$$

$DX = D^3$

$[x = D^2]$ Hence proved

Q.2 Prove that $\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2$

Sol. Row × Column

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \times \begin{vmatrix} -a & -b & -c \\ c & a & b \\ b & c & a \end{vmatrix}$$

$(-(a^3 + b^3 + c^3 - 3abc))\ (-(a^3 + b^3 + c^3 - 3abc))$

$= (a^3 + b^3 + c^3 - 3abc)^2$

Q.3 Prove that $\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 \end{vmatrix} = 2(a_1 - a_2)\,(a_2 - a_3)\,(a_3 - a_1)\,(b_1 - b_2)\,(b_2 - b_3)\,(b_3 - b_1)$

Sol. $\begin{vmatrix} a_1^2 + b_1^2 - 2a_1b_1 & a_1^2 + b_2^2 - 2a_1b_2 & a_1^2 + b_3^2 - 2a_1b_3 \\ a_2^2 + b_1^2 - 2a_2b_1 & a_2^2 + b_2^2 - 2a_2b_2 & a_2^2 + b_3^2 - 2a_2b_3 \\ a_3^2 + b_1^2 - 2a_3b_1 & a_3^2 + b_2^2 - 2a_3b_2 & a_3^3 + b_3^2 - 2a_3b_3 \end{vmatrix}$

$$\begin{vmatrix} a_1^2 & a_1 & 1 \\ a_2^2 & a_2 & 1 \\ a_3^2 & a_3 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ -2b_1 & -2b_2 & -2b_3 \\ b_1^2 & b_2^2 & b_3^2 \end{vmatrix}$$

$C_1 \to C_3$

Take common (–2)

$$1 - 2\begin{vmatrix} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{vmatrix}\begin{vmatrix} 1 & 1 & 1 \\ b_1 & b_2 & b_3 \\ b_1^2 & b_2^2 & b_3^2 \end{vmatrix}$$

$$2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$$

Q.4 Prove that $\begin{vmatrix} a_1\ell_1 + b_1m_1 & a_1\ell_2 + b_1m_2 & a_1\ell_3 + b_1m_3 \\ a_2\ell_1 + b_2m_1 & a_2\ell_2 + b_2m_2 & a_2\ell_3 + b_2m_3 \\ a_3\ell_1 + b_3m_1 & a_3\ell_2 + b_3m_2 & a_3\ell_3 + b_3m_3 \end{vmatrix} = 0$

Sol. Column × Column

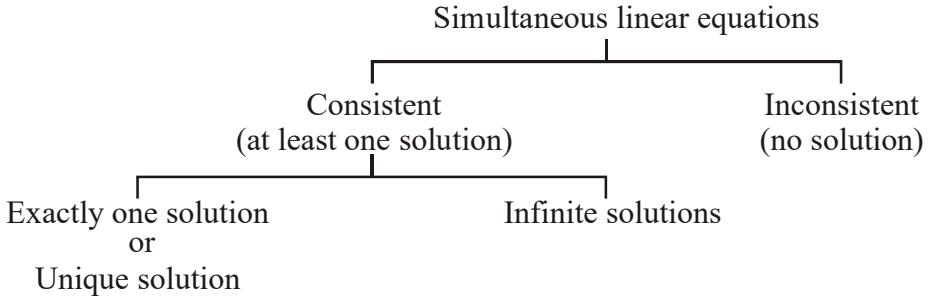
$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ 0 & 0 & 0 \end{vmatrix} \times \begin{vmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

Q.5 Prove that $\begin{vmatrix} 2 & \alpha+\beta+\gamma+\delta & \alpha\beta+\gamma\delta \\ \alpha+\beta+\gamma+\delta & 2(\alpha+\beta)(\gamma+\delta) & \alpha\beta(\delta+\gamma)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) & 2\alpha\beta\gamma\delta \end{vmatrix} = 0$

Sol. Row × Column

$$\begin{vmatrix} 1 & 1 & 0 \\ \gamma+\delta & \alpha+\beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix} \times \begin{vmatrix} 1 & \alpha+\beta & \alpha\beta \\ 1 & \gamma+\delta & \gamma\delta \\ 0 & 0 & 0 \end{vmatrix} = 0$$

4.1 SYSTEM OF EQUATIONS



Solution

Trivial	**Non-Trivial**
(All variables are zero)	(Atleast one variable is nonzero)
Solution corresponding to the point origin.	Solution corresponding to the point other than origin.

(a) Two Equations involving two variables :

Let, $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$ then :

(i) Consistent Equations : Definite & unique solution (Intersecting lines)

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ $\Rightarrow$ Given equations are consistent with unique solution

(ii) Inconsistent Equations : No solution (Parallel lines)

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ $\Rightarrow$ Given equations are inconsistent

(iii) Dependent Equations : Infinite solutions (Identical lines)

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ $\Rightarrow$ Given equations are consistent with infinite solutions

(b) Three equations involving two variables :

$a_1x + b_1y + c_1 = 0$ $\quad a_2x + b_2y + c_2 = 0$ $\quad a_3x = b_3y + c_3 = 0$

4.2 CRAMER'S RULE

Consider three linear simultaneous equation in 'x', 'y', 'z'

$a_1x + b_1y + c_1z = d_1$ ...(i)

$a_2x + b_2y + c_2z = d_2$ ...(ii)

$a_3x + b_3y + c_3z = d_3$ ...(iii)

and

$$\text{if } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \qquad D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \qquad D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

then using Crammer's rule of determinant we get

$$\frac{x}{D_x} = \frac{y}{D_y} = \frac{z}{D_z} = \frac{1}{D}$$

i.e. $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$, $z = \frac{D_z}{D}$

Case-I If $D \neq 0$

Then $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$, $z = \frac{D_z}{D}$

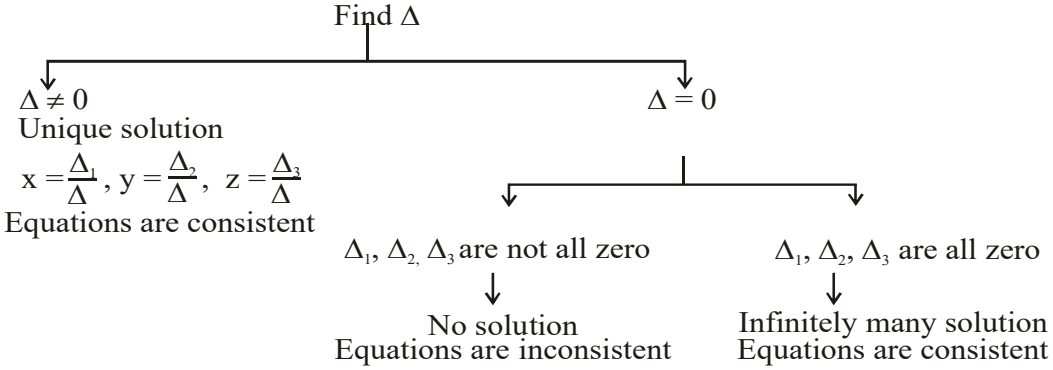
$\therefore$ The system is consistent and has unique solutions

Case-II If $D = 0$ and

(i) If at least one of D_x, D_y or D_z is not zero then the system of equations is inconsistent i.e. has no solution

(ii) If $d_1 = d_2 = d_3 = 0$ or D_x, D_y, D_z are all zero then the system of equations has infinitely many solutions.

The above can be shown diagrammatically as follows



Note: (1) Case of unique solution : These planes will have a unique point of intersection.

(2) Case of infinite solutions : One plane will pass through line of intersection of other two planes.

(3) Case of no solution : Three planes will form a prism.

Q.1 Solve: $2x + y + 2z = 7,\ x + y - z = 0,\ 3x + 4y + 5z = 17$

Ans $x = 1,\ y = 1,\ z = 2$

Sol. $D = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & -1 \\ 3 & 4 & 5 \end{vmatrix}$

$= 2(5 + 4) - 1(5 + 3) + 2(4 - 3)$
$= 18 - 8 + 2$
$= 12$

$D_x = \begin{vmatrix} 7 & 1 & 2 \\ 0 & 1 & -1 \\ 17 & 4 & 5 \end{vmatrix}$

$= 7(5 + 4) - 0 + 17(-1 - 2)$
$= 63 - 51$
$= 12$

$D_y = \begin{vmatrix} 2 & 7 & 2 \\ 1 & 0 & -1 \\ 3 & 17 & 5 \end{vmatrix}$

$= -7(5 + 3) - 0 - 17(-2 - 2)$

$= -56 + 68$

$= 12$

$D_z = \begin{vmatrix} 2 & 1 & 7 \\ 1 & 1 & 0 \\ 3 & 4 & 17 \end{vmatrix}$

$= 7(4 - 3) - 0 + 7(2 - 1)$
$= 24$

$x = \frac{D_x}{D} = \frac{12}{12} = 1$

$y = \frac{D_y}{D} = \frac{12}{12} = 1$

$z = \frac{D_z}{D} = \frac{24}{12} = 2$

Q.2 Solve: $2x + y + z = 5,\ \ 3x + 2y + 4z = 10,\ 7x + 4y + 6z = 20$

Ans. $(x, y, z) \equiv (2\lambda, 5 - 5\lambda, \lambda)$

Sol. $D=\begin{vmatrix}2 & 1 & 1\\3 & 2 & 4\\7 & 4 & 6\end{vmatrix}$

$D = 2(12 - 16) - 1(18 - 28) + 1(12 - 14)$
$-8 + 10 - 2 = 0$

$D_x=\begin{vmatrix}5 & 1 & 1\\10 & 2 & 4\\20 & 4 & 6\end{vmatrix}$

$5(12 - 16) - 1(60 - 80) + 1(40 - 40)$
$5(-4) - 1(-20) + 1(0)$
$= -20 + 20$
$= 0$

$D_y=\begin{vmatrix}2 & 5 & 1\\3 & 10 & 4\\7 & 20 & 6\end{vmatrix}$

$= 2(60 - 80) - 5(18 - 28) + 1(60 - 70)$
$= 2(-20) - 5(-10) + 10$
$= -40 + 50 - 10$
$= 0$

$D_2=\begin{vmatrix}2 & 1 & 5\\3 & 2 & 10\\7 & 4 & 20\end{vmatrix}$

$= 2(40 - 40) - 1(60 - 70) + 5(12 - 14)$
$= 10 - 10$
$= 0$

Solution set :- Let $z = \lambda$

$2x + y = 5 - \lambda$ $\quad (\lambda, 4, 2)$
$3x + 2y = 10 - 4\lambda$ $\quad = (2\lambda, 5 - 5\lambda, \lambda)$

$x = 2\lambda$
$y = 5 - 5\lambda$

Q.3 The system of linear equations $3x - 2y - kz = 10; 2x - 4y - 2z = 6; x + 2y - z = 5m$

is inconsistance then : **[JEE-Main,Feb-2021]**

रैखिय समीकरण निकाय $3x - 2y - kz = 10; 2x - 4y - 2z = 6; x + 2y - z = 5m$

असंगत है, यदि : **[JEE-Main,Feb-2021]**

Ans. $k = 3, m \neq \frac{4}{5}$

Sol. $3x-2y-kz=10$

$2x-4y-2z=6$

$x+2y-z=5m$

eliminate y

$2x+(1-k)z=7$

$4x+(-k-1)z=10+5m$

For inconsistant $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$

$\frac{2}{4}=\frac{1-k}{-k-1}\neq\frac{7}{10+5}$

$k=3$

$m\neq\frac{4}{5}$

Q.4 If the system of equations $kx+y+2z=1$; $3x-y-2z=2$; $-2x-2y-4z=3$ has infinitely many solutions, then k is equal to _____. **[JEE-Main,Feb-2021]**

यदि समीकरण निकाय $kx+y+2z=1$; $3x-y-2z=2$; $-2x-2y-4z=3$ के अनन्त हल हैं, तो k बराबर है : **[JEE-Main,Feb-2021]**

Ans. 21

Sol. Here, $D=Dx=Dy=Dz=0$

$$D=\begin{vmatrix}k & 1 & 2\\ 3 & -1 & -2\\ -2 & -2 & 4\end{vmatrix}=0 \quad k\in R$$

$$Dx=\begin{vmatrix}1 & 1 & 2\\ 2 & -1 & -2\\ 3 & -2 & -4\end{vmatrix}=0$$

$$Dy=\begin{vmatrix}k & 1 & 2\\ 3 & 2 & -2\\ -2 & 3 & -4\end{vmatrix}=0 \Rightarrow k=21$$

$$Dz = \begin{vmatrix} k & 1 & 1 \\ 3 & -1 & 2 \\ -2 & -2 & 3 \end{vmatrix} = 0 \quad k = 21$$

Hence, k = 21

Exception of cramer's rule :-

Case of three parallel planes

$2x + y + 4z = 3$

$2x + y + 4z = 7$

$2x + y + 4z = 10$

$D = 0 = D_x = D_y = D_z$ But it is a case of no solution.

4.3 SYSTEM OF HOMOGENEOUS LINEAR EQUATIONS

Let $a_1x + b_1y + c_1z = 0$ ..... (i)

$a_2x + b_2y + c_2z = 0$ ..... (ii)

$a_3x + b_3y + c_3z = 0$ ..... (iii)

$\Rightarrow \quad D_x = D_y = D_z = 0$

$\therefore$ The system always possesses atleast one solution $x = 0, y = 0, z = 0$, which is called **Trivial** solution, i.e. this system is always **consistent.**

Check value of D

$D \neq 0$: **Unique Trivial** solution

$D = 0$: **Trivial & Non-Trivial** solutions (infinite solutions)

Q.1 $px + (p + 1)y + 2z = 0$

$(p + 1)x + 2y + pz = 0$

$2x + py + (p + 1)z = 0$

find p if system has

(i) infinite solutions (ii) non-trivial solution (iii) non-zero solution

Ans. (i) $p = -3/2$ (ii) $p = -3/2$ (iii) $p = -3/2$

Sol. (i), (ii), (iii) $D = \begin{vmatrix} p & p+1 & 2 \\ p+1 & 2 & p \\ 2 & p & p+1 \end{vmatrix} = 0$

$$p = -\frac{3}{2}$$